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Total Synthesis of a PSGL-1 Glycopeptide Analogue for Targeted Inhibition of P-Selectin

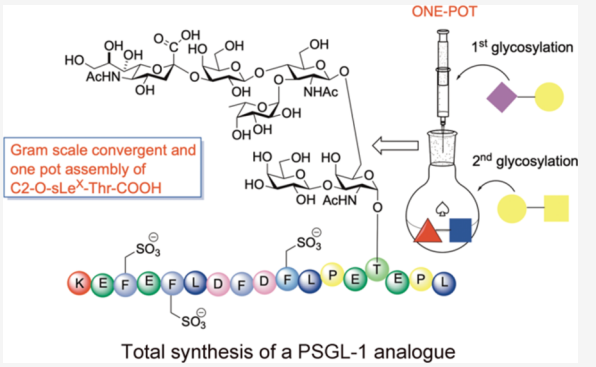
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ABSTRACT: The high affinity interaction between P-selectin glycoprotein ligand-1 (PSGL-1) and P-selectin is mediated by a multimotif glycosulfopeptide (GSP) recognition domain consisting of clustered tyrosine sulfates and a Core 2 *O*-glycan terminated with sialyl Lewis^X (C2-*O*-sLe^X). These distinct GSP motifs are much more common than previously appreciated within a wide variety of functionally important domains involved in protein−protein interactions. However, despite the potential of GSPs to serve as tools for fundamental studies and prospects for drug discovery, their utility has been limited by the absence of chemical schemes for synthesis on scale. Herein, we report the total synthesis of GSnP-6, an analogue of the N-terminal domain of PSGL-1, and potent inhibitor of P-selectin. An efficient, scalable, hydrogenolysis-free synthesis of C2-*O*-sLe^X-Thr-COOH was identified by both convergent and orthogonal one-pot assembly, which afforded this crucial building block, ready for direct use in solid phase peptide synthesis (SPPS). C2-*O*-sLe^X-Thr-COOH was synthesized in 10 steps with an overall yield of 23% from the 4-*O*,5-*N* oxazolidinone thiosialoside donor. This synthesis represents an 80-fold improvement in reaction yield as compared to prior reports, achieving the first gram scale synthesis of SPPS ready C2-*O*-sLe^X-Thr-COOH and enabling the scalable synthesis of GSnP-6 for preclinical evaluation. Significantly, we established that GSnP-6 displays dose-dependent inhibition of venous thrombosis in vivo and inhibits vaso-occlusive events in a human sickle cell disease equivalent microvasculature-on-a-chip system. The insights gained in formulating this design strategy can be broadly applied to the synthesis of a wide variety of biologically important oligosaccharides and *O*-glycan bearing glycopeptides.



1. INTRODUCTION

Many cell−cell recognition processes are dependent on the presentation of unique glycans in the context of other aglycone components, but the combined recognition functions of such determinants are poorly understood. Glycosulfopeptides represent a novel class of such complex glycoconjugates that are present in a wide variety of cell surface receptors, including P-selectin glycoprotein ligand-1 (PSGL-1),[1−4] GPIbα,[5,6] endoglycan,[7,8] and CCR5,[9,10] among other related biomacromolecules. All told, glycosulfopeptides play critical roles in disorders of innate immunity, thrombosis, cancer, viral infections, and ischemia-reperfusion syndrome.[11] Detailed structural and mechanistic studies have revealed the presence of a recognition motif common to many glycosulfopeptides that consist of clustered tyrosine sulfates and a Core 2 *O*-glycan that bears a sialyl Lewis^X-containing hexasaccharide epitope (C2-*O*-sLe^X).[4,11] Despite the potential of glycosulfopeptides to serve as tools for biological studies and their promise as therapeutic agents, their utility has been limited by the absence of chemical schemes for their synthesis on a scale.

As a representative biological molecule with a glycosulfopeptide domain, PSGL-1 is a cell surface glycoprotein that is expressed on all leukocytes and supports leukocyte recruitment in the context of a broad range of inflammatory and immune responses.[12,13] PSGL-1 is a ligand for P-, E-, and L-selectins, but binds with the highest affinity to P-selectin that is expressed on activated platelets and endothelial cells.[1,14,15] Ligation of P-selectin by PSGL-1 constitutes the initial "capture and rolling" step in the leukocyte-endothelial cell adhesion cascade.[13,15,16] Likewise, the interaction of PSGL-1 with P-selectin on activated platelets leads to leukocyte-platelet, platelet−platelet, and tumor-platelet aggregates that promote the adhesion and infiltration of both inflammatory

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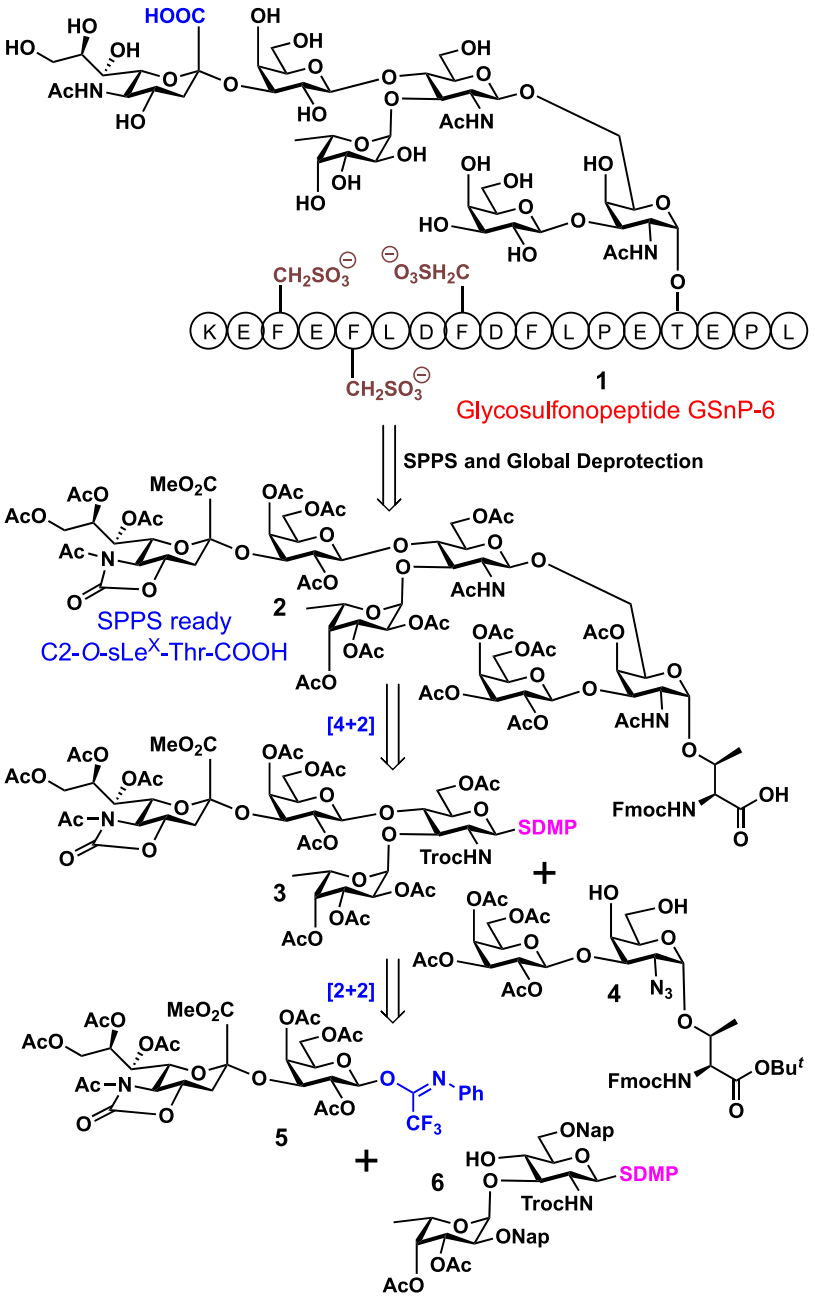


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Scheme 1. Retrosynthetic Analysis



and tumor cells with the generation of metastatic niches.[17−19] The recent approval of an anti-P-selectin antibody for the treatment of sickle cell disease has confirmed that P-selectin is a druggable target for human disease.[20]

We have recently reported a rationally designed novel glycosulfopeptide mimic of N-terminal PSGL-1, GSnP-6, as the first synthetic high affinity (K_d 22 nM) and specific P-selectin antagonist for human therapy.[21] Moreover, we have demonstrated the ability of a pegylated form of GSnP-6 (P-G6) to prevent venous thrombosis after intravenous administration in an established preclinical model of disease without an attendant increase in bleeding risk, as characteristically observed with anticoagulants.[22] Chemoenzymatic synthesis of GSnP-6 was accomplished from a synthetic glycopeptide intermediate, followed by the enzymatic addition of galactose, sialic acid, and fucose residues using respective glycotransferases and nucleotide donors. Enzymatic synthesis offers great value in the context of protecting group free synthesis but has a number of limitations including a requirement for costly sugar nucleotide donors and recombinant enzymes and a need for bioreactor process development and downstream processing. Moreover, while glycosyltransferases are highly selective, they are associated with prolonged reaction times and often express poorly in *Escherichi coli*. As a consequence, glycosyltransferases have been widely used for lab-scale preparation of complex carbohydrates,[1,21,23] but are not commonly used for large-scale production processes.

In principle, total chemical synthesis should provide an expedient route to C2-*O*-sLe^X as a common building block for glycosulfopeptide mimics of PSGL-1, as well as many other biologically important glycopeptides. However, significant limitations inherent to all reported schemes have hindered

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large-scale synthesis of both an appropriately derivatized sLe^X[24−33] intermediate and the related C2-*O*-glycan hexasaccharide bearing amino acid.[34−36] Characteristic of the few reported synthetic routes to achieve these important intermediates, Kunz et al. described a 15-step scheme to synthesize a C2-*O*-sLe^X hexasaccharide ready for solid phase peptide synthesis (SPPS) on milligram scale with a mere overall yield of 0.29%.[35]

Herein, we report a practical, scalable, hydrogenolysis-free, and SPPS compatible assembly of C2-*O*-sLe^X-Thr-COOH. Key synthetic challenges were addressed through the design of disaccharide building blocks accessible through a highly efficient stereoselective route, in a minimum number of steps, on a multigram scale, compatible with convergent and orthogonal one-pot strategies. The sLe^X derivative ready for [4 + 2] glycosylation with a Core-1-diol was synthesized in >6 g quantity with an overall yield of 48% in 6 steps from a sialic acid 4-*O*, 5-*N*-oxazolidinone thiosialoside donor. We also developed a novel stereoselective scheme that led to multigram (>10 g) scale synthesis of the Core-1-diol. The overall yield for the SPPS ready C2-*O*-sLe^X-Thr-COOH glycoamino acid starting from the oxazolidinone thiosialoside donor was 23% in 10 total steps, with synthesis performed on a scale exceeding 1 g. As a consequence of using a low-temperature NIS/TfOH activation strategy, this synthetic approach does not suffer from any regioselective loss at the [4 + 2] glycosylation step with yields as high as 88%, and as the intermediate is not subject to basic conditions of prolonged exposure to palladium hydroxide during hydrogenolysis, the principal Fmoc group remains intact. This synthesis represents an 80-fold improvement in reaction yield compared to prior reports, achieving the first gram scale synthesis of SPPS ready C2-*O*-sLe^X-Thr-COOH. Significantly, we also demonstrate that C2-*O*-sLe^X-Thr-COOH can be synthesized through an orthogonal one-pot scheme. Solid phase peptide synthesis using the C2-*O*-sLe^X-Thr-COOH building block afforded GSnP-6 through a fragment-condensation strategy, followed by late-stage deprotection of sulfonate and glycan protecting groups. These innovations enabled scalable synthesis of a long-acting pegylated variant of GSnP-6, P-G6, for further preclinical evaluation. P-G6 demonstrated a favorable pharmacokinetic profile following subcutaneous administration, with dose-dependent reduction of thrombus formation in a murine model of venous thrombosis at drug plasma concentrations consistent with the IC_{50} for the P-selectin blockade. Moreover, evaluation of GSnP-6 in a microvasculature-on-a-chip system confirmed the ability of this drug candidate to markedly inhibit vaso-occlusive events among individuals with sickle cell disease already maintained on hydroxyurea as a conventional medical therapy.

2. RESULTS

2.1. Retrosynthetic Analysis. The synthesis of glycosulfonopeptide GSnP-6 **1** requires the generation of C2-*O*-sLe^X-Thr-COOH **2** on a multigram scale suitable for solid phase peptide synthesis. A retrosynthetic analysis identified three structural building blocks necessary to achieve this goal, including Core-1-diol **4**, peracetylated sialyl galactose donor **5**, and fucosylated glucosamine acceptor **6** (Scheme 1). We envisioned the synthesis of C2-*O*-sLe^X-Thr-COOH **2** from a regioselective [4 + 2] glycosylation between sLe^X tetrasaccharide **3** and a Core-1-diol **4**. To address the shortcomings of prior strategies, we postulated that the sLe^X tetrasaccharide **3** would be best obtained through a [2 + 2] glycosylation between a peracetylated sialylated galactose donor **5** and a thio-2,6-dimethylphenyl (SDMP) aglycone bearing fucosylated galactosamine acceptor **6**. This would afford a sLe^X tetrasaccharide **3** bearing thioglycoside prepared for [4 + 2] glycosylation. Moreover, we postulated that these building blocks would facilitate assembly of the threonine hexasaccharide through either convergent block synthesis [2 + 2] + 2 or through an orthogonal one-pot strategy, enabling multigram synthesis (Scheme 1).

A number of key elements were incorporated within the design of the synthetic scheme. First, an acetate (OAc) protected galactose acceptor was used for sialylation, which facilitated the use of milder basic conditions during global deprotection in order to avoid β-elimination. This strategy provided access to the sialylated galactose *N*-phenyl acetimidate donor **5** on a multigram scale in a minimum number of steps and avoided the need for further functional group modifications. Second, being electron deficient, the peracetylated sialylated galactose *N*-phenyl acetimidate disaccharide donor provided access to multigram quantities of conjugation ready sLe^X tetrasaccharide **3** by suppressing orthoester formation,[37] Chapmann byproduct,[38] α-glycoside formation, as well as acyl migration from donor to acceptor,[39,40] which are common challenges during β-galactosylation, especially in the context of large-scale reaction conditions. Third, an optimally functionalized fucosylated glucosamine C4−OH acceptor, a building block recognized as less reactive and sterically less accessible due to C3-*O*-fucosylation, was identified through a [2 + 2] parallel screening approach. This building block was able to avoid undesirable side reactions, including aglycone transfer reactions[41] and the loss of an acid labile fucose under Bronsted acidic conditions, which together often limit the ability to achieve higher yields and effective separation of the desired product. Finally, the Core-1-diol was accessed in a stereoselective fashion using di-*tert*-butylsilylene group (DTBS)-mediated galactosamination[42,43] of a threonine acceptor with almost exclusive formation of the α-anomer on multigram scale, followed by galactosylation and removal of the DTBS group.

2.2. Synthesis of Disaccharide Building Blocks. *2.2.1. Synthesis of an α-Sialylated Galactose Donor 5.* Stereoselective chemical synthesis of α-sialosides remains a significant challenge in carbohydrate chemistry and has been thoroughly explored over several decades.[44,45] Obstacles include the electron withdrawing nature of a carboxylate group linked to a tertiary anomeric center, absence of a participating group at the C3 position, E_1 elimination leading to a glycal byproduct, and an oxocarbenium ion favoring the formation of a β-sialoside. Prior investigations have evaluated modifications of the leaving group at C2,[46−49] functionalization of the acetamide group at C5,[50−52] use of auxiliary groups at C1 and C3,[53,54] in addition to the use of a variety of promoters for the sialylation reaction.[55]

Ultimately, the identification of an appropriate leaving group at C2 in combination with a highly electron withdrawing protecting group at N5 position led Takahashi,[56] Crich,[52] and De Meo[57] to develop α-selective 4-*O*,5-*N*-oxazolidinone and *N*-acetyl-4-*O*,5-*N*-oxazolidinone thiosialoside donors. Nonetheless, the low reactivity of thiosialoside donors has limited their use in programmable, orthogonal one-pot synthesis, motivating Wong et al.[58] to employ a dibutyl phosphate leaving group with successful use of a highly reactive, α-selective phosphate donor in one-pot synthesis of various sialic

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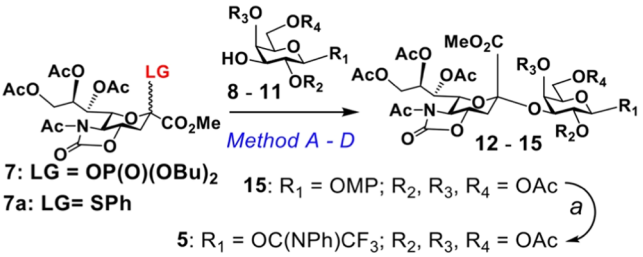
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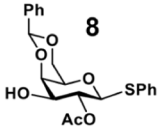
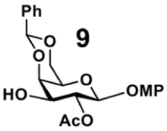
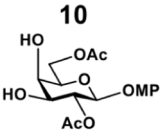
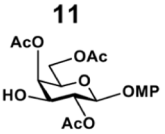
acid containing oligosaccharides. Ando et al. have recently reported that macrobicyclized, constrained sialic acid donors are also able to achieve excellent yields with complete α-selectivity, with wide applicability to a variety of acceptors and complex sialoside targets.[59] In the context of our target, the rational design of a suitably functionalized acceptor was crucial to achieve α-sialylation on a multigram scale and minimize subsequent steps to obtain sialylated galactose donor **5** for use in either convergent or orthogonal one-pot schemes. Most disarmed galactose acceptors that lead to high yielding α-sialylation incorporate benzoyl ester (OBz) protecting groups.[60] However, the use of OBz-based building blocks proved to be incompatible with glycopeptide synthesis due to β-elimination of the entire glycan in the presence of basic conditions required for global deprotection (unpublished data). Therefore, the suitability of acceptors **8**−**11** bearing acetate (OAc) protecting groups was evaluated under a variety of reaction conditions using sialic acid phosphate donor **7** (Table 1).

Sialylation using acceptor **8** at −78 °C in CH_2Cl_2 led to disaccharide **12** in 42% yield with 16% recovery of an aglycone transfer byproduct (Table 1, entry 1).[41] Sialylation with acceptor **9**, conducted under two different conditions in CH_2Cl_2, achieved greater selectivity and higher yield when acetonitrile was used as a cosolvent leading to disaccharide **13** (Table 1, entry 2). The diol acceptor **10** was subjected to sialylation in a CH_2Cl_2/CH_3CN [2:1] solvent system at −50 °C and produced disaccharide **14** in 62% yield with excellent selectivity (Table 1, entry 3). Sialylation of the disarmed acceptor **11** under similar solvent and temperature conditions as that used for the diol acceptor generated disaccharide **15** with the exclusive α-anomer in 59% yield (Table 1, entry 4). Significantly, disarmed acceptor **11** required the fewest protecting group manipulations to produce a peracetylated disaccharide donor ready for glycosylation. Further optimization, by modifying the ratio of CH_2Cl_2/CH_3CN from [2:1] to [1:10], together with an increase in reaction temperature to −35 °C, afforded disaccharide **15** in 84% yield with exclusive α-selectivity (Table 1, entry 4). We speculate that the increase in yield is due to acetonitrile stabilization of the oxocarbenium ion intermediate, in addition to the higher reactivity of acceptor **11** at −35 °C. Sialylation using donor **7a** and acceptor **11** using NIS and TfOH afforded the desired disaccharide **11** in 83% yield with exclusive α-selectivity (Table 1, entry 4), avoiding the extra step for phosphate donor synthesis. The *p*-methoxy phenyl (OMP) protected disaccharide **15** was converted to the *N*-phenyl trifluoroacetimidate donor **5** in 68% yield over two steps. To our knowledge, this is the first systematic study of sialylation using acetate (OAc) protected galactose acceptors and represents the shortest route for the synthesis of peracetylated sialyl galactose disaccharide donor **5** suitable for glycosylation on a multigram scale.

2.2.2. Synthesis of Disaccharide Acceptors 4 and 6. Three fucosylated glucosamine disaccharide acceptors **21**, **22**, and **6** were synthesized by [1 + 1] glycosylation, followed by regioselective opening of the naphthyl (Nap) acetal protecting group in a one-pot scheme (Scheme 2). The 4,6-Nap protected glucosamine C3−OH acceptors, **16**, **17**, and **18**, displaying 3′-*O*-*t*-butyldiphenylsilyl (OTBDPS), thiophenyl (SPh), and SDMP aglycones, respectively, were synthesized from D-glucosamine·HCl in six or fewer steps in good overall yield. Commencing with L-fucose, donors **19** and **20** were prepared in nine or fewer steps with overall yields of 54 and

Table 1. Stereoselective α-Sialylation[d]



Entry	Acceptor	Method[a]	Product: yield[b] [α:β][c]
1		A	**12**: 42% [α-only]
2		B C	**13**: 56% [8:1] **13**: 64% [10:1]
3		C	**14**: 62% [α-only]
4		C D	**15**: 59% [α-only] **15**: 84% [with **7**, α-only] **15**: 83% [with **7a**, α-only]

[a]Sialylation attempted using donor **7** [1 equiv], acceptors **8**−**11** [1.2 equiv], TMSOTf [1 equiv] and activated 4 Å MS using methods A−D. For **7a** NIS [2 equiv], TfOH [0.2 equiv] and method D was used. Method A [CH_2Cl_2, −78 °C, 2 h], Method B [CH_2Cl_2, −50 °C, 1 h], Method C [CH_2Cl_2:CH_3CN (2:1), −50 °C, 1 h], Method D [CH_2Cl_2:CH_3CN (1:10), − 35 °C, 0.5 h]. [b]Isolated yields. [c]α:β ratio from crude NMR. [d]Reagents and Conditions: [a] i. CAN, CH_3CN:H_2O [4:1], 0 °C, 3 h ii. $CF_3C(NPh)Cl$, Cs_2CO_3, 0 °C, 2 h, 68% over two steps.

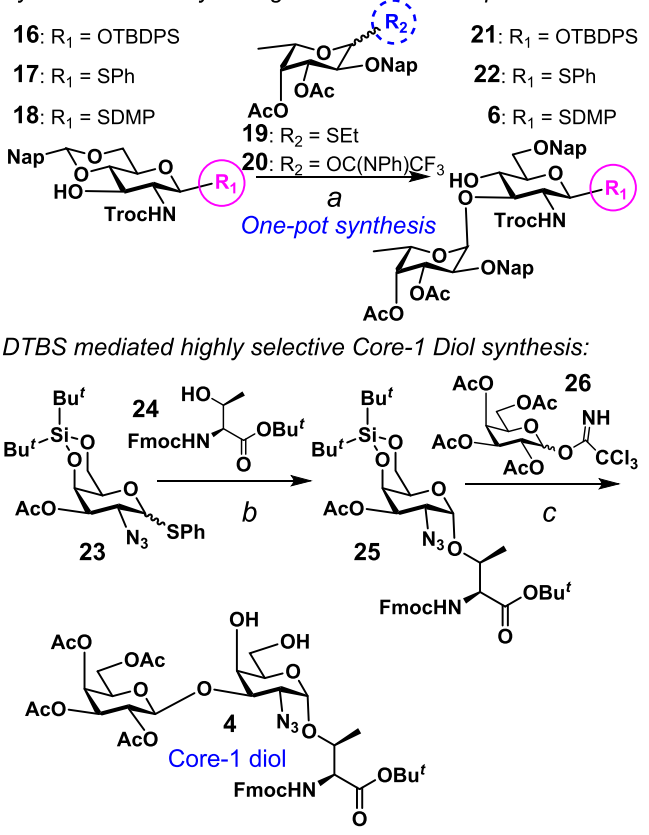
36%, respectively. The glucosamine acceptor **16** bearing an OTBDPS aglycone was coupled to donor **19** under NIS, TfOH activation conditions, and selective ring opening afforded **21** in 71% yield. Glucosamine acceptors **17** and **18**, bearing SPh and SDMP aglycones, respectively, were coupled to donor **20** using TMSOTf as a promoter, followed by selective ring opening, which afforded **22** and **6**, with yields of 64 and 61%, respectively. Stereoselectivity of the α-fucosidic linkage was confirmed using one-dimensional (1D) and two-dimensional (2D) NMR, with exclusive formation of the α-isomer.

The Core-1-diol acceptor **4** was synthesized on a multigram scale (Scheme 2). The di-*tert*-butylsilylene (DTBS) group favors stereoselective α-galactosylation by stabilization of the oxocarbenium ion via through-space electron donation, while limiting β-facial nucleophilic attack due to the presence of a

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Scheme 2. Synthesis of Disaccharide Building Blocks[a]



[a]Reagents and Conditions: [a] i. **19**, NIS, TfOH, −30 °C, 0.5 h ii. Et_3SiH, TfOH, CH_2Cl_2, −78 °C, 5 h, 71%, for **21** over two steps or i. **20**, 0 °C TMSOTf, 0.5 h ii. Et_3SiH, TfOH, CH_2Cl_2, 5 h, −78 °C, 64% for **22** and 61% for **6** over two steps. [b] **24**, NIS, TfOH, 0 °C, 1 h, 81%. [c] i. $N_2H_4{\cdot}H_2O$, MeOH, rt, 3 h, 91% ii. **26**, TMSOTf, 0 °C, 0.5 h, 79% iii. TBAHF, THF, rt, 2 h, 86%.

bulky *tert* butyl group.[42,43] As a consequence, stereoselective attack from the α-face of the oxocarbenium ion leads to the diastereoselective formation of an α-glycosidic linkage. As such, the DTBS-protected donor **23** was coupled to the NHFmoc, *tert* butyl protected threonine acceptor **24** under NIS and TfOH activation conditions at 0 °C. The threonine bearing galactosamine was synthesized with excellent α-selectivity [α:β = 32:1] in 81% yield. Hydrazine-mediated deacetylation of compound **25** afforded a threonine bearing galactosamine acceptor intermediate in 91% yield, which was then coupled to peracetylated galactose trichloroacetimidate donor **26** to produce the DTBS-protected Core-1 disaccharide in 79% yield. Deprotection of the DTBS group using tributylamine hydrogen fluoride (TBAHF) at room temperature afforded the desired Core-1 diol in 86% yield.

2.3. Multigram Scale Convergent Assembly of C2-*O*-sLe^X-Thr-COOH Suitable for Solid Phase Peptide Synthesis. Three distinct aglycones bearing fucosylated glucosamine acceptors **21**, **22**, and **6** were screened in a [2 + 2] glycosylation reaction with sialylated galactose donor **5** to provide the desired sLe^X tetrasaccharide (Scheme 3). The [2 + 2] coupling of disaccharide donor **5** with disaccharide acceptor **21** bearing an OTBDPS aglycone was performed using TMSOTf as a promoter at 0 °C and afforded the OTBDPS bearing tetrasaccharide **27** in 68% yield. The [2 + 2] glycosylation of disaccharide acceptor **22** bearing an SPh aglycone with donor **5** using TMSOTf as a promoter at 0 °C afforded the SPh bearing tetrasaccharide **28** in 52% yield, but with an aglycone transfer byproduct isolated in 15% yield.[41]

In an attempt to suppress the aglycone transfer side reaction, glycosylation was performed at −40 °C with varying donor and acceptor equivalents, but the byproduct continued to be observed. As a consequence, [2 + 2] glycosylation was subsequently performed with donor **5** and disaccharide acceptor **6** bearing an SDMP aglycone at 0 °C using TMSOTf as a promoter. The SDMP bearing sLe^X tetrasaccharide **29** was obtained in 77% yield without any trace of aglycone transfer byproduct, consistent with the ability of 2,6-dimethylphenyl (DMP) to effectively block aglycone transfer with thioglycosides.[61] Significantly, neither orthoester formation[37] nor acyl migration from donor to acceptor[39,40] were observed, which was attributed to the highly disarmed nature of the donor and emphasizes the importance of rationally designed building blocks to circumvent side reactions that hinder higher glycosylation yields and facilitate product purification. The presence of a β-glycosidic linkage was confirmed by 1D and 2D NMR. Naphthylmethyl (Nap) protecting groups were oxidatively cleaved using DDQ[62] in aqueous CH_2Cl_2 and acetylated by reaction with acetic anhydride in pyridine to afford peracetylated sLe^X **3** and **31**, bearing SPh and SDMP aglycones in 92 and 71% yields, respectively. This approach facilitated multigram synthesis of a sLe^X glycan, which can be readily conjugated to a variety of linkers at the reducing end, thereby providing a means to generate large quantities of target compounds.

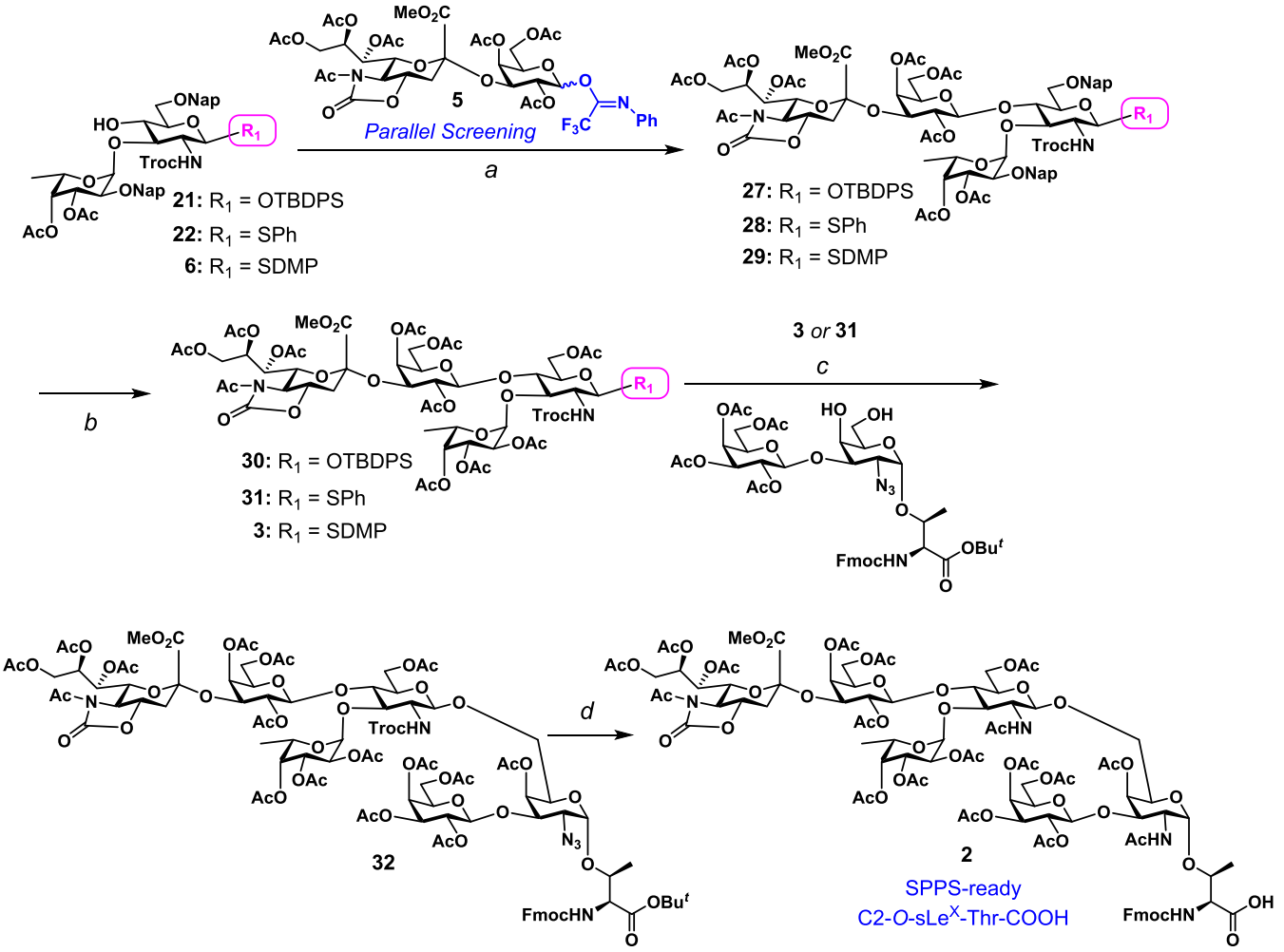
Peracetylated sLe^X tetrasaccharide donors **3** and **31** were coupled to Core-1-diol **4** under NIS, TfOH activation conditions at 0 °C, which was followed by acetylation of C4−OH providing hexasaccharide **32** from **3** or **31** in 88 and 69% yields, respectively. Notably, the good reactivity of disarmed peracetylated sLe^X tetrasaccharides was observed under NIS activation conditions at 0 °C. 1D and 2D NMR confirmed regioselectivity and exclusive formation of a β-glycosidic linkage due to participation of the NHTroc group at the C2 position of glucosamine. Conversion of the NHTroc and N_3 protecting groups to acetamides in hexasaccharide **32** using Zn/AcOH/Ac_2O[63] followed by cleavage of the *tert* butyl ester group of threonine afforded C2-*O*-sLe^X-Thr-COOH **2** in 55% yield over two steps (Scheme 3). Significantly, the overall yield for SPPS ready C2-*O*-sLe^X-Thr-COOH starting from an oxazolidinone thiosialoside donor was 23% in 10 total steps, which afforded 1.27 g of the target compound in a single setting.

2.4. Orthogonal One-Pot Strategy. Disaccharide building blocks **4a**, **5**, and **6** were designed to be compatible with either convergent [2 + 2] + 2 synthesis or an orthogonal one-pot strategy to afford C2-*O*-sLe^X-Thr-COOH **2**. Screening of various parameters, including temperature, ratio of building block equivalents, and normal or inverse glycosylation approaches, led to the successful synthesis of the desired hexasaccharide in very good yield on a 250 mg scale (Scheme 4). The one-pot glycosylation reaction utilizes two simultaneous glycosylation sequences based on disaccharide building blocks **4a**, **5**, and **6**. In the optimized protocol, the first glycosylation follows an inverse protocol, in contrast to the convergent scheme, with the fucosylated glucosamine acceptor **6** serving as the rate-limiting reagent and donor **5** added dropwise at −78 °C. The in situ generated naphthylmethyl

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Scheme 3. Multigram Scale Convergent Synthesis of C2-*O*-sLe^X-Thr-COOH[a]



[a]Reagents and Conditions: [a] i. **5** [1 equiv], TMSOTf [0.15 mol %], 0 °C, 0.5 h, 68% with for **27** [1.5 equiv], 52% for **28** [1.5 equiv] and 77% for **29** [1.5 equiv]. [b] DDQ, aq. CH_2Cl_2, 0 °C to rt, 10 h iii. Ac_2O, pyridine, DMAP, rt, 6 h, 74% for **30** over two steps, 71% for **31** over two steps, 92% for **3** over two steps. [c] i. **3** [1 equiv] or **28** [1 equiv], NIS [2 equiv], TfOH [0.2 mol %], 0 °C, 1 h ii. Ac_2O, DMAP, pyridine, 69% from **31** over two steps and 88% from **3** over two steps. [d] i. Zn:AcOH:Ac_2O:THF, 0 °C to rt, 4 h, 73% ii. TFA:CH_2Cl_2, 5 h, 75% for **2**.

containing protected sLe^X was then subjected to a second glycosylation reaction with Core-1 acceptor **4a** to generate the naphthylmethyl containing protected hexasaccharide **33**. Final deprotection sequences furnished C2-*O*-sLe^X-Thr-COOH **2** ready for solid phase peptide synthesis.

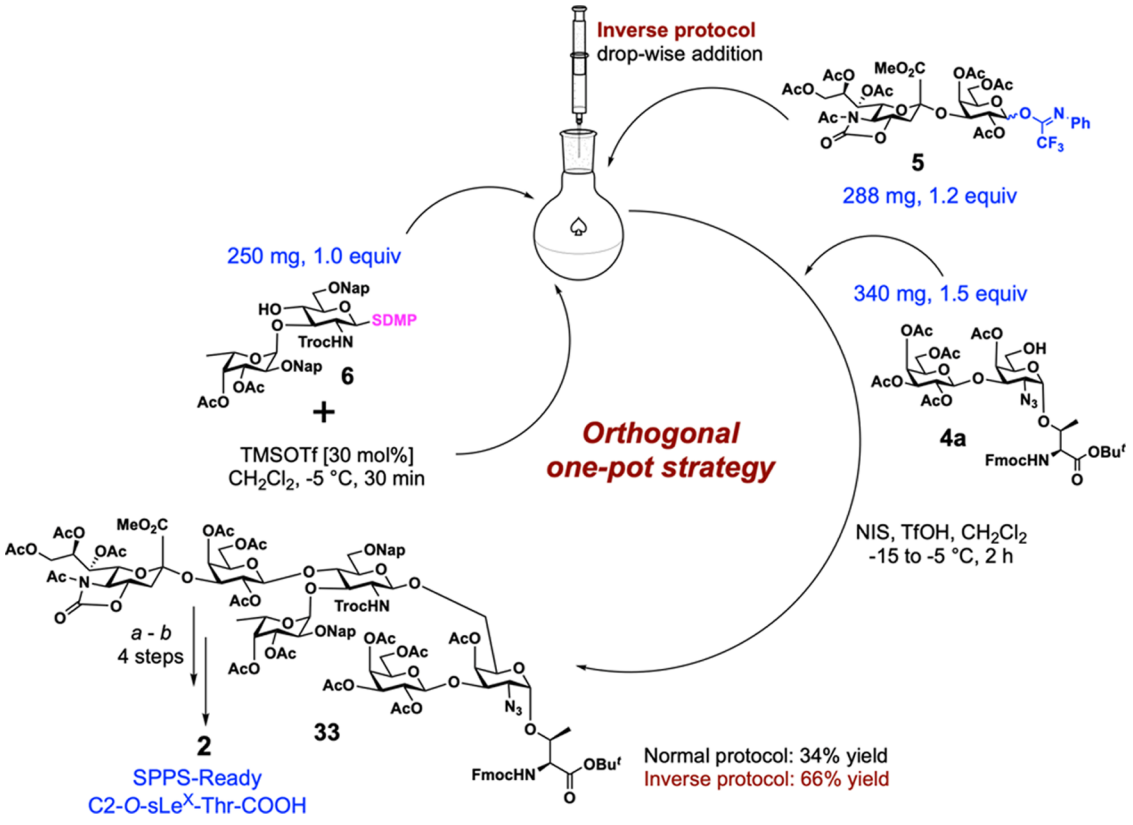
Initially, one-pot glycosylation was attempted under a standard glycosylation protocol with donor **5** and acceptor **6**, followed by the addition of acceptor **4a**. Due to the low reactivity of fucosylated C4−OH glucosamine acceptor **6**, nearly half of the acceptor remained unreacted at −5 °C after 30 min (Table 2, entry 1). After the addition of the Core-1 acceptor **4a** under NIS/TfOH activation conditions, the formation of various side products resulted in a yield of 27%. To address the low reactivity of **6**, excess donor **5** (>2.5 to 3 equiv) would have been required under normal glycosylation conditions. To mitigate this requirement, a one-pot reaction was performed at lower temperature with extended reaction times and slow warming of the reaction mixture. Conducting the reaction at −78 to 0 °C with 1.5 (Table 2, entry 2) and 2 equiv (Table 2, **entry 3**) of donor **5** for a period of 4 h led to yields of 30 to 34%.

Due to the low reactivity of acceptor **6**, we postulated that an inverse glycosylation protocol with dropwise addition of donor **5** to a reaction mixture of acceptor **6** and TMSOTf (30 mol %) would shift the equilibrium toward the formation of the desired sLe^X tetrasaccharide **29**. The inverse glycosylation protocol was attempted with 2 equiv of donor **5** (Table 2, entry 4) and 1 equiv of acceptor **6**, leading to the total consumption of the fucosylated glucosamine acceptor in less than 30 min at −5 °C. As a consequence of using excess acceptor (2 equiv), the desired sLe^X was observed as the major product and also led to the formation of a hemiacetal. Addition of acceptor **4a** under NIS and TfOH activation led to the formation of desired hexasaccharide **33** in 52% yield. The inverse protocol was further optimized to minimize formation of a hemiacetal and tertiary butyl group cleavage under acidic glycosylation conditions. Use of 1.2 equiv of donor **5** under the inverse protocol led to clean conversion of acceptor **6** to sLe^X, as demonstrated by thin layer chromatography (TLC) and matrix-assisted laser desorption/ionization (MALDI) within 15 min. The acceptor **4a** was then added, followed by addition of NIS/TfOH at −10 °C with slow warming to 0 °C, resulting in the formation of the desired hexasaccharide **33** in 66% yield

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Scheme 4. Orthogonal One-Pot Synthesis[a]



[a]Reagents and Conditions: [i] **5** [1.2 equiv], **6** [1 equiv], TMSOTf [30 mol %], 4 Å MS activated, − 5 °C, 0.5 h [ii] i. **4** [1.5 equiv], NIS [2 equiv + 1 equiv after 30 min], TfOH [10 mol %], 2 h, −15 to −5 °C, 66% for **33**. [a] i. DDQ, CH_2Cl_2:H_2O (20:1), 12 h. ii. Pyridine, Ac_2O, DMAP, 8 h, 72% over two steps [b]. i. Zn:AcOH:Ac_2O:THF, 0 °C to rt, 4 h ii. TFA:CH_2Cl_2, 3 h or TIPS:H_2O:TFA, 5 h, 0 °C to rt, 62% for **2** over two steps.

Table 2. Reaction Screening for Orthogonal One-Pot Synthesis of Hexasaccharide 33

entry	compound **6**/**5**/**4a** (equiv)	glycosylation protocol	temp (°C) [2 + 2]/[4 + 2]	time (h) [2 + 2]/[4 + 2]	yield[c] (%)
1	1.0/1.5/1.5	normal[a]	−5/−15 to −5	0.5/2	27
2	1.0/1.5/1.5	normal	−78 to 0/−15 to −5	4/2	31
3	1.0/2.0/1.5	normal	−78 to 0/−15 to −5	4/2	34
4	1.0/2.0/1.5	inverse[b]	−5/−15 to −5	0.5/2	52
5	1.0/1.2/1.5	inverse	−5/−15 to −5	0.5/2	66

[a]Both donor **5** and acceptor **6** were dried and activation was done using TMSOTf, followed by addition of acceptor **4a** under NIS/TfOH activation [see SI]. [b]Donor **5** was added to a solution of acceptor **6** and 30 mol % TMSOTf, followed by addition of acceptor **4a** under NIS/TfOH activation conditions [see SI]. [c]Isolated yield after column chromatography.

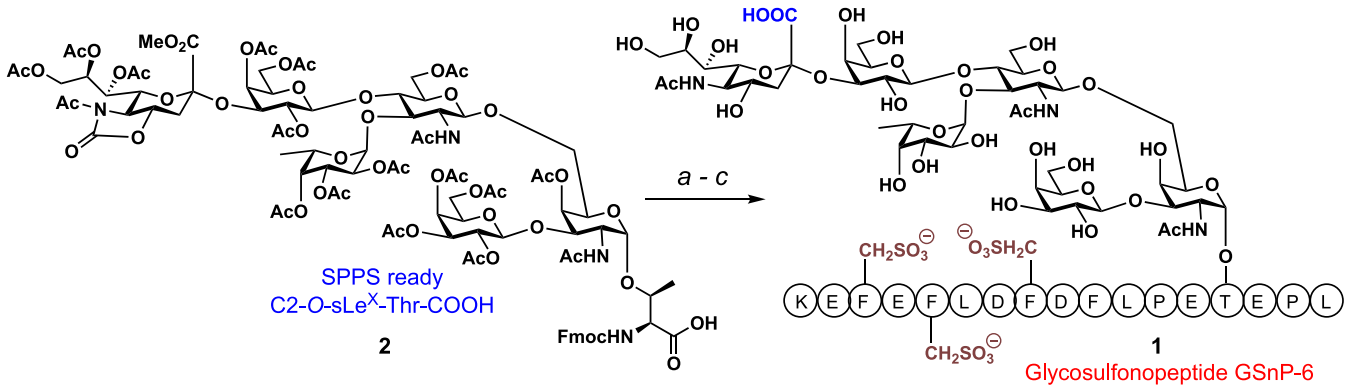
at a 250 mg scale (Table 2, entry 5). Hexasaccharide **33** was subjected to DDQ oxidation conditions to remove the Nap-protecting groups,[59] followed by acetylation to provide the peracetylated hexasaccharide **32** in 72% yield over two steps. Compound **32** was subject to Zn/AcOH-mediated acetamide formation[60] and *tert* butyl ester cleavage using TFA/Et_3SiH, which afforded C2-*O*-sLe^X-Thr-COOH **2** in 62% yield, ready for solid phase peptide synthesis (Scheme 3).

2.5. Solid Phase Peptide Synthesis of GSnP-6 and Global Deprotection. Solid phase peptide synthesis was accomplished on a 70 μmol scale via a fragment-condensation strategy. Common peptide synthesis reagents were used including NovaSyn TGA resin, Fmoc-amino acids, 1-hydroxy benzotriazole (HOBt), 2-(1*H*-benzotriazol-1-yl)−1,1,3,3-tetramethyluronium hexafluorophosphate (HBTU), and 20% piperidine for Fmoc deprotection. Acid labile tyrosine sulfate building blocks were replaced with a bioisosteric sulfonate analogue, Fmoc Phe (p-CH_2SO_3H). In brief, Fmoc-Glu-OH and Fmoc-Pro-OH were coupled to NovaSyn TGA resin preloaded with Fmoc-Leu-OH, followed by coupling of the Fmoc-hexasaccharide-Thr-OH **2** in the presence of O-(benzotriazol-1-yl)-*N*,*N*,*N*′,*N*′-tetramethyluronium tetrafluoroborate (TBTU), HOBT, and trimethylpyridine in DMF (Supporting Information). To deprotect the Fmoc group, the resin was rinsed with 2% DBU and 1% dithiothreitol (DTT) in DMF, which also led to the opening of the 4-O,5-N-oxazolidinone ring of sialic acid. In order to maximize yield, the N-terminal fragment of GSnP-6 was then coupled directly to the resin. The fully protected 13 residue N-terminal fragment was synthesized separately on 2-CTC resin preloaded with Fmoc-Glu-OH in 63% yield. The synthesized peptide fragment displayed >90% purity on high-performance liquid chromatography (HPLC) and was used without further purification. After washing, the fragment was coupled in a

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Scheme 5. Solid Phase Peptide Synthesis and Global Deprotection of GSnP-6[a]



[a]Reagents and conditions: [a] Fmoc SPPS via fragment-condensation strategy, NovaSyn TGA resin followed by 95% aq. TFA:Et_3SiH:EDT. [b] H_2/Pd−C. [c] 0.1 M NaOMe, aq. LiOH.

solution of TBTU, HOBT, and trimethylpyridine in DMF. The final product was cleaved off the resin using aqueous 95% trifluoracetic acid (TFA), 1,2-ethanedithiol (EDT), and triethyl silane, precipitated by cold ether and purified by reversed-phase high-performance liquid chromatography (RP-HPLC) to provide 122 mg of the protected glycosulfonopeptide in 38% overall yield.

Trichloroethyl (TCE)-protected phenylalanine sulfonates were initially deprotected by hydrogenation using palladium on carbon, and the product purified by RP-HPLC with a yield of 76%. Acetate protecting groups and the methyl ester group were subsequently hydrolyzed using sodium methoxide at pH 8.0−8.5, followed by lithium hydroxide at pH 10.5−11, affording the fully deprotected glycosulfonopeptide, GSnP-6 (**1**), after RP-HPLC purification in 41% yield (Scheme 5).

2.6. GSnP-6 Inhibits Sickle Cell Microvascular Occlusion. Blockade of the P-selectin/PSGL-1 pathway has been reported to reduce the frequency of vaso-occlusive crises among individuals with sickle cell disease (SCD).[20] We evaluated the ability of GSnP-6 to inhibit vaso-occlusive events using a human SCD equivalent microvasculature-on-a-chip system. This lab-on-a-chip device is a poly-(dimethylsiloxane) (PDMS)-based platform composed of a microfluidic network of 32 branching microchannels lined with a confluent layer of human endothelial cells. Whole blood is perfused through this network of 30 μm diameter channels at 1 μL/min, which simulates the wall shear stress (1−4 dyn/cm^2) and diameter of postcapillary venules (Figure 1A,B).[64] The system enables quantitative assessment of vaso-occlusion with high spatial resolution, and importantly, the interactions among blood cells, endothelial cells, and soluble factors that represent the underlying pathophysiology of SCD responsible for microvascular occlusion are reproduced. Significantly, the platform has enabled the evaluation of potential therapies in a physiologically relevant setting using whole blood from individuals with SCD.

Hydroxyurea is a commonly used drug to prevent the complications of SCD by decreasing sickle hemoglobin polymerization within erythrocytes. However, despite the use of this drug, the frequency of sickle cell crises remains high. Blood from children (ages 2 to 18 years) with SCD (n = 10), already maintained on hydroxyurea, was recalcified and perfused for 25 min through the lab-on-chip system in the presence or absence of GSnP-6 (120 μM). Immediately following perfusion, devices were imaged, and occlusion was quantified using iClots, an open-source stand-alone software that uses predefined workflows based upon well-validated image processing and algorithms to measure the percentage of occluded microchannels.[65] Treatment with GSnP-6 led to a marked reduction of vaso-occlusive events (p < 0.009) (Figure 1C). The degree to which thrombosis was attenuated in a given individual by GSnP-6 was unrelated to the measured percentage of hemoglobin S or the total concentration of hemoglobin (Supporting Figure 1).

2.7. P-G6 Displays Potent Antithrombotic Activity after Subcutaneous Administration. Subcutaneous (SC) drug delivery is a desirable route for patient self-administered dosing. To extend the half-life of GSnP-6, PEGylation of the N-terminus was performed, as previously described, using a 40 kDa methoxy poly(ethylene glycol)-succinimidyl valerate to afford **P-G6**.[22] Plasma concentration−time data were collected in mice after subcutaneous administration of **P-G6** at 2 μmol/kg and pharmacokinetic parameters defined, including a $t_{1/2}$ of 19.5 ± 0.8 h, which facilitated daily drug dosing (Figure 1D and Table 3).

The ability of **P-G6** to mediate thromboprophylaxis in the setting of once daily subcutaneous drug administration was evaluated in a murine model of nonocclusive venous thrombosis. In an initial study, **P-G6** was administered subcutaneously, the vena cava subjected to an electrolytic injury 4 h later, and a second dose administered 24 h after the initial dose. The thrombus was harvested 48 h after the initial drug dose. Dose-dependent inhibition of venous thrombosis was confirmed with a reduction in thrombus formation consistent with that achieved by enoxaparin, a low molecular weight heparin administered at a clinically relevant dose (Figure 1E). In a subsequent trial, **P-G6** was administered subcutaneously once daily, and the vena cava injured 4 h after the 4th dose (76 h) with thrombus harvested 24 h after a fifth and final subcutaneous dose. As plasma drug concentrations approached equilibrium, effective thrombus inhibition was achieved at a lower daily dose of 1 μmol/kg. **P-G6** plasma concentrations at study termination were measured and were consistent with IC_{50} values previously reported for blockade of P-selectin binding to murine and human leukocytes (Figure 1F).[22]

3. DISCUSSION

An efficient and effective strategy for the scalable synthesis of C2-*O*-sLe^X as a crucial intermediate for many glycopeptides of

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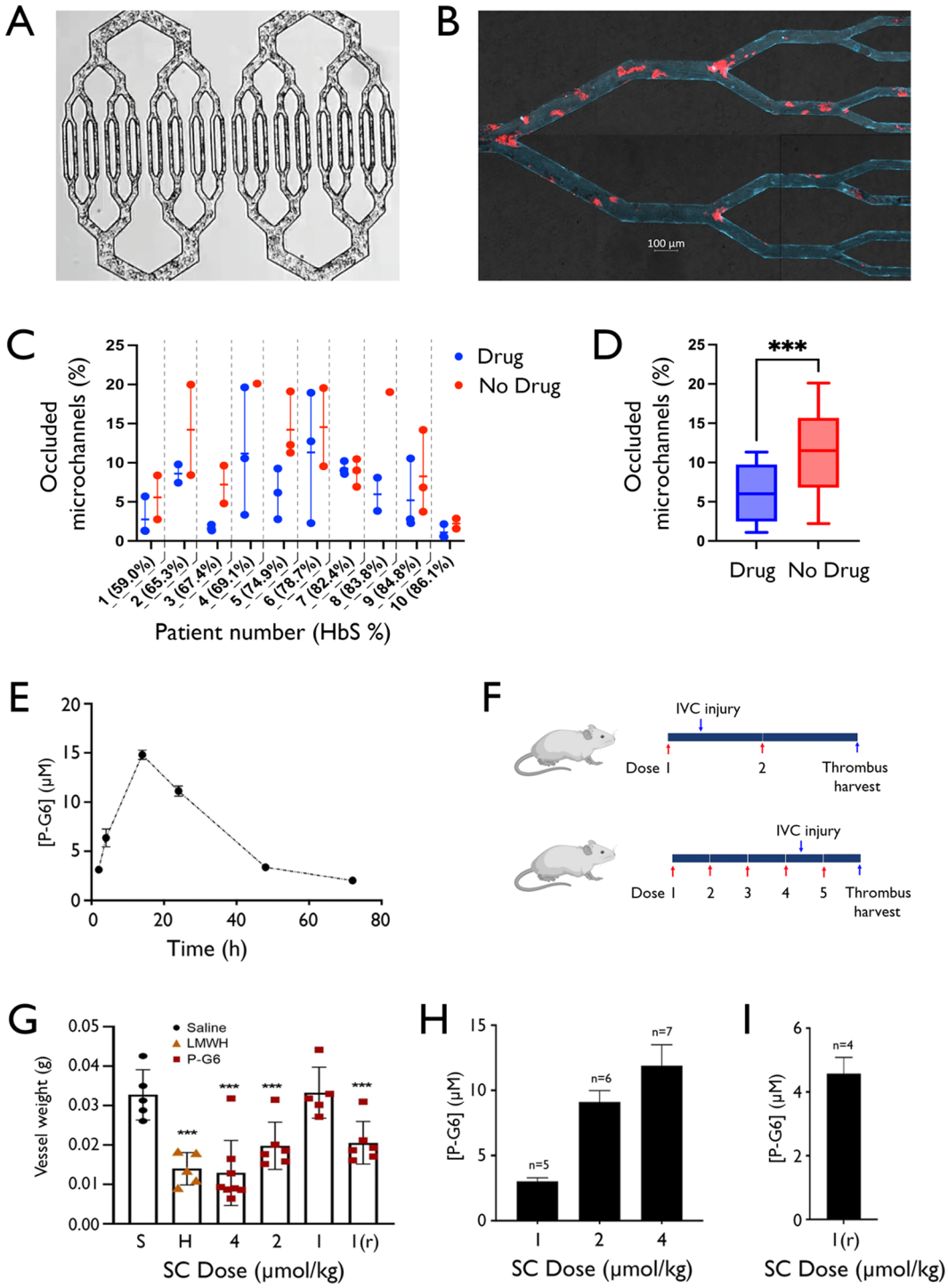


Figure 1. (A) A microvasculature lab-on-system consisting of 32 microchannels lined with human endothelial cells was used to monitor occlusive events in whole blood from individuals with SCD (HbS) maintained on hydroxyurea therapy. (B) Occlusive events were quantified through automated image analysis of aggregates in the microchannels by measuring CD41 fluorescent area. (C) Blood samples from individuals with SCD (n = 10) were perfused through "endothelialized" microvasculature-on-chip devices with or without **GSnP-6** (120 μM), and microchannel occlusion was monitored over time. The percentage of occluded microchannels of each individual device is represented by the dots, and the bar represents the average across multiple runs using the same patient sample. (D) Perfusion of SCD blood samples with **GSnP-6** (120 μM) led to significantly lower levels of microchannel occlusion compared to perfusion of SCD blood samples without the drug across the entire cohort of patients (p = 0.009). (E) Plasma concentration profile of **P-G6** following subcutaneous administration at 2 μmol/kg (mean ± SEM, n > 3 mice/time point). Calculated pharmacokinetic parameters include $t_{1/2\beta}$ 19.5 ± 0.8 h, C_{max} 640.7 ± 20.5 (mg/L), and T_{max} 14.0 ± 0.0 h. (F) Thrombus weight was measured after subcutaneous administration of saline (S), enoxaparin (H; 6 mg/kg), or **PG-6** at 1, 2, or 4 μmol/kg, administered 4 h before and 24 h after vena cava injury. A second trial was conducted in which **P-G6** was administered subcutaneously once daily at 1 μmol/kg (1(r)) with vena cava injury occurring 4 h after 4th dose and thrombus weight determined 24 h after the fifth and final dose. (G) Subcutaneous administration of **PG-6** led to a dose-dependent reduction in venous thrombosis (***p < 0.001). **P-G6** plasma concentrations were measured at study termination following both (**H**) 2-dose and (**I**) 5-dose administration protocols.

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Table 3. Pharmacokinetic Parameters of P-G6

parameter	P-G6 (SC)
C_{max} (mg/dL)	640.7 ± 20.5
T_{max} (h)	14.0 ± 0.0
$t_{1/2\beta}$ (h)	19.5 ± 0.8
AUClast/dose (mg·h/L)/(mg/kg)	243.5 ± 10.0
AUC0-INF/dose (mg·h/L)/(mg/kg)	272.5 ± 13.8

interest has not been achieved due to multiple shortcomings of existing schemes. In particular, inefficient sialylation strategies have been associated with poor stereo- and regioselectivity using diol or triol acceptors.[24,25] Aglycone transfer side reactions often occur during glycosylation[41] and Chapmann rearrangement byproducts[38] are known to limit β-galactosylation. The synthesis of the C2-*O*-glycan hexasaccharide has been limited by poor regioselectivity of the β-glycosylation reaction, which is required to achieve efficient [4 + 2] coupling.[34,35] Moreover, stereoselective synthesis of the Core-1-diol on a multigram scale has been difficult to accomplish because of challenges related to β-galactosylation. Further, incompatibility of protecting groups has often led to loss of fucose and cleavage of *O*-glycosidic linkages under acidic conditions required for deprotection of the Fmoc group,[66,67] which is necessary to incorporate the *O*-glycan-bearing amino acid into a peptide synthesis protocol. Likewise, the frequent requirement for late-stage post glycosylation modifications, including hydrogenolysis for removal of benzyl protecting groups, reintroduction of an Fmoc group, and global acetylation, have also contributed to low product yield.[35] Finally, even after the synthesis of the protected glycopeptide, basic saponification conditions required for global deprotection of the glycan under harsher conditions risks β-elimination and loss of the entire glycan.[68] All told, access to glycosylation ready disaccharides or tetrasaccharide donors requires replacing protecting groups at multiple stages throughout the course of the synthesis, extending the number of steps of the reaction sequence with associated product losses. In the process, reaction scalability and overall yield are significantly compromised and the generation of adequate quantities of C2-*O*-sLe^X bearing target glycopeptides necessary for fundamental biological investigations as well as preclinical studies rendered all but inaccessible.[34−36,69]

A total chemical synthesis is described for C2-*O*-sLe^X-Thr-COOH, which is suitable for incorporation within a solid phase peptide synthesis scheme without further modification. As an example of one application, this critical intermediate afforded the total chemical synthesis of GSnP-6, a glycosulfonopeptide analogue of the N-terminal domain of PSGL-1, which has been identified as a potent inhibitor of P-selectin and facilitates the synthesis of a variety of other *O*-glycan bearing peptides and glycosulfopeptides for fundamental biological studies as well as the design of therapeutic and diagnostic agents. Through the use of three disaccharide building blocks with compatible protecting groups, we designed both convergent and one-pot schemes for the scalable synthesis of the hexasaccharide bearing amino acid. Beginning with an oxazolidinone thiosialoside donor, a sialyl Lewis^X derivative was initially synthesized on a scale exceeding 6 g in 6 steps with an overall yield of 48%. This afforded C2-*O*-sLe^X-Thr-COOH on a scale exceeding 1 g in 4 additional steps ready for SPPS with an overall yield of 23%.

Several significant innovations were critical to improving efficiency and providing an 80-fold improvement in reaction yield as compared to prior reports,[35] achieving the first gram scale synthesis of SPPS ready C2-*O*-sLe^X-Thr-COOH. In particular, use of benzyl free peracetylated building blocks established an expedient, practical, and scalable route to a hexasaccharide bearing amino acid without the need for hydrogenolyis. Synthesis of the sialyl Lewis^X derivative benefited from the use of a peracetylated sialyated galactose *N*-phenyl acetimidate disaccharide donor, which suppressed orthoester formation, Chapmann rearrangement byproducts, and α-glycoside formation. Moreover, exploiting a thio-2,6-dimethylphenyl (SDMP) donor group was crucial to avoiding aglycone transfer, and incorporation of naphthylmethyl protecting groups further enhanced the stability of the [2 + 2] glycosylation reaction. Finally, employing a di-*tert*-butylsilylene (DTBS) group proved beneficial in the synthesis of the Core-1-diol by favoring stereoselective α-galactosylation, which provided access to a single glycosylated isomer. Total chemical synthesis of GSnP-6 was successfully achieved by a peptide fragment-condensation protocol, and by avoiding the use of benzoyl-based building blocks, β-elimination of the entire glycan was averted during the late-stage basic deprotection sequences. Scalable synthesis of GSnP-6 and a long-acting pegylated variant, P-G6, enabled further preclinical evaluation, which confirmed potent inhibition of thrombus formation in a murine model of venous thrombosis after subcutaneous administration as well as a marked reduction of vaso-occlusive events in a validated human microvasculature-on-chip model of sickle cell disease.

To date, synthetic P-selectin inhibitors, which have been largely designed to mimic the sLe^X moiety of PSGL-1, have displayed limited clinical efficacy, presumably due to low binding affinity to P-selectin. For example, among the most promising candidates, rivipansel (GMI-1070) demonstrated initial clinical benefit in a phase 2 trial of sickle cell pain crisis, but its low activity (IC_{50} 423 μM) required infusion of approximately 2 g of drug per day and a later phase 3 trial failed to meet primary study end points.[70,71] Likewise, bimosiamose (TBC-1269) advanced to clinical trials directed at psoriasis, asthma, and chronic obstructive pulmonary disease.[72] However, only marginal benefit was observed, likely as a consequence of modest inhibitory activity exemplified by high micromolar inhibitory activity.[73] The discovery that quinoline salicylic acids inhibit P-selectin led to the development of PSI-697 and PSI-421 as the first orally available P-selectin inhibitors. Nonetheless, both of these compounds have been associated with modest inhibitory activity (IC_{50} ∼ 200 μM), consistent with weak binding affinity and have not been further advanced to clinical trials.[74,75] The inherent challenge of designing small molecule inhibitors of the P-selectin/PSGL-1 pathway is due to the presence of large and relatively flat surfaces with extensive contact areas and multiple interaction points that characterize the interface between P-selectin and PSGL-1, along with the absence of a deep and well-defined binding pocket.

In this regard, GSnP-6 is representative of an important class of glycopeptide therapeutics for effectively targeting such interactions with several advantages over inhibitory small molecules and antibodies. Specifically, as a glycopeptide mimic of the critical structural motifs of PSGL-1, GSnP-6 fits into and blocks the interaction interface with high specificity and affinity with a reduced likelihood of off-target effects. Moreover,

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GSnP-6 is amenable to further modification to enhance the stability and bioavailability. In addition, as compared with larger protein-based therapies, such as monoclonal antibodies, glycopeptide drugs have a lower potential for immunogenicity, which is particularly important for long-term chronic therapy.

In summary, longstanding challenges that have limited the ability to achieve a scalable synthetic pathway for an essential *O*-glycan bearing an amino acid were overcome through the rational design of disaccharide building blocks. The assembly of these building blocks afforded hydrogenolysis-free synthesis of C2-*O*-sLe^X-Thr at the gram scale, ready for solid phase peptide synthesis, which led to the total synthesis of GSnP-6, a high affinity P-selectin inhibitor with promising preclinical properties. It is particularly noteworthy that targeting the P-selectin/PSGL-1 pathway using a therapeutic antibody has been recently validated as a treatment for sickle cell disease. However, sickle cell disease is a global public health problem with the majority of the 300,000 affected children born annually located in sub-Saharan Africa.[76] Without intervention, 50% or more will die before 5 years of age.[77] From the perspective of cost of goods, supply, and logistics, including storage as a lyophilized powder, a synthetic inhibitor that can be self-administered represents a particularly important therapeutic option for underserved and impoverished regions of the world lacking technology intensive healthcare infrastructure. We believe that the insights gained in formulating a design strategy for C2-*O*-sLe^X-Thr will be broadly applicable to the synthesis of a wide variety of biologically important oligosaccharides and *O*-glycan bearing glycopeptides.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c05090.

Full experimental details; ^{1}H and ^{13}C NMR spectra of all new compounds (PDF)

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Notes

The authors declare no competing financial interest.

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